

The Human Conversion Imperative

Machine Acceleration, Human Potential, and the Reversibility Window in an AI-Robotic World

A standalone theory proposal for converting machine capability into durable human agency before dependency locks in

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Status K0 theory proposal and empirical research agenda. This paper does not claim that a global “bad mode” has already occurred, that AI capability grows at one universal rate, that industrial robot installations equal robot capability, or that human potential can be represented by one validated scalar. Current 2024–2026 evidence is used only to establish acceleration, feedback, learning-risk, concentration, and embodied-automation conditions that make prospective measurement rational. The core proposal is conditional and falsifiable: when machine capability grows faster than institutions convert it into broadly retained human agency, ownership, exit, world-answerability, and social reproduction, technological progress can increase human dependency even while assisted performance and aggregate output rise.

Central thesis. The central problem of an AI-rich society is not whether machines become capable. It is whether machine capability is converted into durable human capability and broadly distributed systemic standing at a rate sufficient to prevent dependency, deskilling, gate concentration, and institutional lock-in. The paper therefore replaces a one-dimensional “AI helps or harms” question with a vector-valued Human Conversion architecture and adds a Reversibility Principle: the faster machine capability grows relative to human conversion, and the more costly lost skills, exit routes, ownership structures, and institutions become to restore, the stronger the case for early safeguards that preserve options before crisis is visible.

Edition note. This is a v1.1 glosa-checked edition of the founder’s v1.0 (6 September 2026), which remains the anchor record of this work. Changes from v1.0: (i) citation lineage repair — self-citations [1], [3], [4] now carry their deposited Zenodo DOI and version string, [5] now cites the specific deposited record (Zenodo 22331922) instead of an ambiguous title-only entry, and [2] (*After Labour*) is cited as a companion paper deposited in the same batch, with its claim softened from a stronger verb to “proposes” pending independent verification of the derivation; (ii) one priority-language word removed from the Abstract; (iii) explicit [DEFINITION]/[OPEN] tier tags added next to the defining equations and Propositions 1–4; (iv) one AI-product-name mention in body prose generalised to a generic term (the cited work’s own title keeps its original wording verbatim). No section, equation, table, or reference was added or removed, and no claim was strengthened.

ABSTRACT

Artificial intelligence can improve immediate human performance without necessarily increasing durable human capability. At the same time, AI and robotics can raise aggregate productive capacity while weakening labour-based claims, broad ownership, credible exit, and social standing. Existing literatures typically study these effects separately: human-AI augmentation and synergy, cognitive offloading and learning, automation and labour markets, market concentration, path dependence, or political lock-in. This paper proposes a single bridge concept: Human Conversion. Machine capability is socially beneficial in the strong human-development sense only to the extent that it is converted into retained human capacities and conditions of agency rather than merely substituted for them.

The paper develops four contributions. First, it defines a Human Conversion Vector spanning effective material claim, credible exit, live possibility, corrigible agency, independent epistemic routes, world-answerability, delayed unaided Human Return, social reproduction, and recognized standing. Second, it defines conversion elasticities that permit dimen-

sions to diverge: AI may raise income while lowering skill retention or exit, so a scalar “augmentation” score can conceal managed dependence. Third, it replaces linear collapse narratives with a Bad-Mode State Space and a Reversibility Window grounded in increasing returns, path dependence, and irreversibility. Fourth, it derives an Urgency Vector in which safeguard intensity rises with machine-human conversion gaps, lock-in, severity, and recovery time. Current evidence motivates but does not validate the full model: global AI compute capacity has grown rapidly, industrial robot deployment is heavily concentrated in China (295,000 units installed in 2024, 54% of global installations), several AI-infrastructure layers are highly concentrated, repeated AI interaction can alter later human judgment, and field experiments show that unguarded AI assistance can raise assisted performance while reducing subsequent unaided learning. The paper ends with six minimum safeguard invariants and a falsification programme designed to distinguish productive augmentation from human systemic weakening.

Keywords: human-AI collaboration; human potential; cognitive offloading; agency; Human Return; automation; robotics; path dependence; lock-in; reversibility; political economy; AI

governance.

1 THE MISSING DEPENDENT VARIABLE

The contemporary AI debate asks whether systems increase productivity, improve accuracy, displace jobs, create synergy, or concentrate economic power. These are important outcomes, but none answers the question that becomes decisive under rapid automation: what returns to humans?

The programme paper *When AI Expands Human Potential* defines human potential in an epistemic register: not throughput or convenience, but the capacity to become more corrigible, more world-responsive, less captive to self-fixation, and more capable of carrying truth-pressure into judgment and life [Lahtee, 2026a]. AI can widen alternatives and reflective friction, but candidate generation is not validation; disciplined revision still requires resistant conditions and partially independent routes.

The later *After Labour* paper adds a systemic constraint. Even if AI expands an individual's inquiry, that capacity is not securely realizable if the person lacks effective material claim, credible exit, live alternatives, retained unaided capability, social reproduction, or standing. It proposes that aggregate output can rise while Human Systemic Position falls [Lahtee, 2026b]. The two papers therefore generate a stronger joint problem:

$$\text{Machine expansion} \not\Rightarrow \text{Human expansion.} \quad (1)$$

The missing dependent variable is the rate and direction by which machine capability is converted into durable human capability and systemic position.

2 WHY THIS IS A TIME-SENSITIVE PROBLEM RATHER THAN A DISTANT AGI QUESTION

The urgency argument does not depend on predicting AGI. It depends on observing three conditions simultaneously: fast machine-side change, plausible human/institutional substitution or dependence, and mechanisms of costly reversal.

Stanford's 2026 AI Index estimates that global AI compute capacity grew approximately 3.3-fold per year from 2022, while benchmark performance continued to advance quickly in several reasoning and software-engineering domains [Stanford Institute for Human-Centered AI, 2026]. This is infrastructure and measured capability growth, not a knowledge-growth rate. The distinction is load-bearing: $K_{\text{like}} \neq K_{\text{validated}}$.

Embodied automation is also scaling. The International Federation of Robotics reports 295,000 industrial robots installed in China in 2024, 54% of global installations, with operational stock exceeding two million units [International Federation of Robotics, 2025]. This does not mean robot capability grows at the installation rate, but it shows that cognitive automation is increasingly coupled to physical production at industrial scale.

The social system is not neutral to such acceleration. OECD documents high concentration in several AI-infrastructure lay-

ers: in three segments one firm is reported above 80% global share, and in three others the largest three exceed 60% [OECD, 2025]. Yet concentration is not domination by definition; it becomes structurally important when consequential transitions lack substitutes, interoperability, or credible exit.

Finally, human effects can accumulate before macroeconomic crisis is visible. Glickman and Sharot show in experiments with 1,401 participants that repeated human-AI interaction can alter later perceptual, emotional, and social judgments, amplifying bias [Glickman and Sharot, 2025]. Bastani et al. show that generative-AI access substantially improved practice performance in high-school mathematics, while unrestricted AI use led to worse subsequent unaided performance after access was removed; guardrailed tutoring largely mitigated the harm [Bastani et al., 2025]. A randomized trial in higher education likewise reports lower delayed knowledge retention for unrestricted generative-AI-assisted study [Barcaui, 2025]. Recent reviews conclude that cognitive offloading can support performance but may impede skill acquisition or produce task-specific deskilling depending on use conditions [Cash et al., 2026, Noorbehhahani and Oyibo, 2026].

These findings do not establish global decline. They establish that performance, learning, judgment, and dependence can move in different directions and therefore must be measured before irreversible institutional commitments are made.

3 LITERATURE BOUNDARY: WHAT IS ALREADY KNOWN

This proposal does not claim priority for augmentation, cognitive offloading, path dependence, or automation inequality. Its contribution is the bridge architecture joining them.

Human-AI decision research has long shown that assistance quality depends on task, trust, coordination, explanation, and human uptake. Vaccaro, Almaatouq, and Malone's meta-analysis distinguishes augmentation (human+AI better than human alone) from synergy (human+AI better than both human and AI alone), finding that synergy is not automatic and can be negative in decision tasks [Vaccaro et al., 2024]. Bućinca et al. explicitly argue that AI-assisted systems should optimize human-centric outcomes beyond accuracy, including human learning [Bućinca et al., 2024].

The cognitive-offloading literature provides a complementary warning. External tools can free cognitive resources, but offloading higher-order reasoning can change acquisition, retention, and self-regulation. Recent reviews emphasize that the effects depend on the type and timing of assistance rather than on AI use per se [Cash et al., 2026, Seung and Basham, 2026, Noorbehhahani and Oyibo, 2026]. These literatures support the distinction between dependent offloading and assistance that preserves autonomous competence.

The political-economy literature supplies the macro-side. Automation can displace tasks, create new tasks, alter factor shares, and change the distribution of income and wealth [Autor, 2015, Acemoglu and Restrepo, 2018, Korinek and

[Stiglitz, 2019]. *After Labour* extends that line by separating output from broad claims, ownership from transition control, access from credible exit, and productive necessity from social necessity [Lahtee, 2026b].

Finally, lock-in is not a metaphor invented for AI. Arthur shows that increasing returns can lock technologies into historically contingent paths that become hard to reverse [Arthur, 1989]. Pierson develops the same logic for political institutions, emphasizing sequencing, increasing returns, and critical junctures [Pierson, 2000]. Arrow and Fisher's analysis of uncertainty and irreversibility establishes the general policy logic that preserving options can carry value when future consequences are uncertain and some choices are hard to undo [Arrow and Fisher, 1974].

The residual gap is therefore precise: existing work does not provide a single architecture for asking whether rapid machine capability is being converted into durable, broadly distributed human agency before skill, ownership, exit, and institutions become costly to restore.

4 NON-COLLAPSE DISCIPLINE

The proposal begins with explicit separations:

- AI capability $\neq$ validated knowledge, (2)
- assisted performance $\neq$ human learning, (3)
- augmentation $\neq$ synergy, (4)
- formal options $\neq$ live possibilities, (5)
- access $\neq$ credible exit, (6)
- income transfer $\neq$ future ownership, (7)
- material security $\neq$ agency, (8)
- market concentration $\neq$ domination, (9)
- productive necessity $\neq$ social necessity. (10)

Any theory that collapses these objects can report apparent progress while missing human contraction in another layer.

5 HUMAN CONVERSION AS A VECTOR, NOT A SCALAR

Let M_t denote a declared, normalized measure of machine-side capability relevant to a study. It may combine cost-adjusted AI capability, deployed compute, and embodied machine capacity, but it must never be interpreted as validated knowledge without an independent validation term.

Define the Human Conversion Vector [DEFINITION]

$$C_t^H = \langle \Gamma_t^{\text{eff}}, X_t^H, \Lambda_{H,t}^{\text{live}}, A_{H,t}^{\text{corr}}, R_{H,t}^{\text{route}}, W_{H,t}^{\text{world}}, r_{H,t}, H_t^{\text{cap}}, S_{H,t} \rangle. \quad (11)$$

The components are:

- Γ_t^{eff} : effective material claim after unavoidable rent/debt burdens;
- X_t^H : credible exit or usable substitution away from a consequential AI/robotic gate;

- $\Lambda_{H,t}^{\text{live}}$: breadth and corrigibility of practically live alternatives;
- $A_{H,t}^{\text{corr}}$: capacity to read relevant differences, accept/refuse, act, and revise;
- $R_{H,t}^{\text{route}}$: availability of partially independent critical or verification routes;
- $W_{H,t}^{\text{world}}$: answerability to resistant target-domain conditions;
- $r_{H,t}$: delayed unaided Human Return after decisive AI support is removed;
- H_t^{cap} : social-reproductive human capability stock supported by care, health, education, and community;
- $S_{H,t}$: recognized standing and socially meaningful roles.

These are declared readouts, not substances. No universal weighting is assumed.

For dimension j , define a local Human Conversion Elasticity [DEFINITION]

$$\eta_{j,t}^{HC} = \frac{\partial \ln C_{j,t}^H}{\partial \ln M_t}. \quad (12)$$

This immediately permits mixed trajectories. A deployment can satisfy

$$\eta_{\Gamma}^{HC} > 0, \quad \eta_{r_H}^{HC} < 0, \quad \eta_X^{HC} < 0, \quad (13)$$

meaning that machine capacity raises material consumption while reducing retained human competence and exit. Such a system is not accurately described by one positive “augmentation” score.

Proposition 1 (Conversion non-identity) [OPEN]. An increase in M_t or current human-AI performance is insufficient to establish human expansion unless at least one predeclared Human Conversion dimension improves without unacceptable deterioration in another load-bearing dimension.

6 THE EPISTEMIC CONVERSION MECHANISM

At the micro level, the conversion problem inherits the distinction between closure and corrigibility from *When AI Expands Human Potential*. A high-conversion interaction has the form

$$\begin{aligned} \text{H problem} &\rightarrow \text{AI divergence} \rightarrow \text{H resistance} \\ &\rightarrow \text{World test} \rightarrow \text{H integration} \\ &\rightarrow \text{AI removal} \rightarrow \text{H return.} \end{aligned} \quad (14)$$

A low-conversion interaction can instead be

$$\text{H problem} \rightarrow \text{AI answer} \rightarrow \text{use} \rightarrow \text{dependence.} \quad (15)$$

The point is not that direct AI answers are always harmful. It is that systems optimized only for short-run completion can rationally substitute for the very cognitive work needed to build future independent competence.

The key role separation is asymmetric. AI has comparative advantages in breadth, search, translation, recombination, simulation, and candidate generation. Humans must

Table 1: Empirical anchors motivating the proposal; none validates the full Human Conversion model.

Observed domain	Current evidence	What it licenses
Machine acceleration	Global AI compute capacity estimated at $3.3\times$ annual growth since 2022; frontier benchmark gains remain rapid [Stanford Institute for Human-Centered AI, 2026].	Prospective monitoring of conversion speed; not a claim that knowledge grows $3.3\times$ annually.
Embodied automation	China installed 295,000 industrial robots in 2024 (54% of world installations); operational stock exceeds two million [International Federation of Robotics, 2025].	Physical automation is already scaling; installations are not per-robot capability.
Infrastructure concentration	Several AI hardware/cloud segments exhibit high reported concentration [OECD, 2025].	Gate-risk deserves measurement; concentration alone does not prove domination.
Human-AI feedback	Repeated biased AI interaction altered later human judgments in $n = 1,401$ experiments [Glickman and Sharot, 2025].	Human state after AI interaction can differ from baseline.
Assisted performance vs learning	AI tutoring raised practice performance, while unguarded use reduced later unaided performance in a field experiment [Bastani et al., 2025].	Assisted performance must be separated from durable human return.
Human-AI synergy	Meta-analysis finds augmentation is common but strong synergy is not guaranteed and varies by task [Vaccaro et al., 2024].	“Human+AI” is not intrinsically superior to the best component.

retain problem ownership, normative responsibility, integration, and contact with world-side constraints; criticism and alternative generation can be shared. This predicts that the optimal assistance policy is not “maximum AI use” but task- and learner-contingent support that maximizes conversion rather than substitution.

Proposition 2 (Assistance-conversion divergence) [OPEN]. The AI assistance intensity that maximizes current performance need not maximize delayed Human Return or corrigible agency.

7 BAD MODE IS A STATE SPACE, NOT A STAIR-CASE

Collapse narratives often assume one universal sequence: convenience, deskilling, job loss, ownership concentration, and political lock-in. The architecture rejects that sequence as too strong. Cognitive dependence may occur before labour displacement; informational concentration can precede market concentration; coercive lock-in can arise without private AI ownership.

Define a Bad-Mode State Vector [DEFINITION]

$$B_t = \langle D_t, \bar{G}_t^{\text{conv}}, C_t^{\text{info}}, 1 - X_t^H, 1 - r_{H,t}, 1 - \Lambda_{H,t}^{\text{live}}, 1 - A_{H,t}^{\text{corr}}, 1 - \Gamma_t^{\text{eff}}, 1 - H_t^{\text{cap}} \rangle. \quad (16)$$

D is dependency, $\bar{G}^{\text{conv}}$ conversion-gate power, and C^{info} concentration of consequential information mediation. A “bad mode” is not a point threshold fixed for all societies. It is a sustained trajectory toward a high-risk region in multiple dimensions, especially where the dynamics exhibit positive feedback and costly reversal.

This formulation allows distinct bad trajectories: deskilling without unemployment, material abundance without agency,

cross-border rent extraction without domestic frontier AI, or informational lock-in without industrial automation.

8 REVERSIBILITY, HYSTERESIS, AND THE INTERVENTION WINDOW

Technological and political systems can display increasing returns. Adoption creates complementary investment, habits, data, standards, networks, organizational routines, and political constituencies. Arthur’s technology model and Pierson’s institutional analysis show why early advantages can become self-reinforcing and difficult to reverse [Arthur, 1989, Pierson, 2000].

For each human-conversion dimension j , define recovery cost $C_{j,t}^{\text{rec}}$ and expected recovery time $\tau_{j,t}^{\text{rec}}$. The Reversibility Window [DEFINITION] is the set of times for which restoration remains feasible under declared bounds:

$$W_j = \{t : C_{j,t}^{\text{rec}} \leq \bar{C}_j \wedge \tau_{j,t}^{\text{rec}} \leq \bar{\tau}_j\}. \quad (17)$$

As skills atrophy, standards harden, ownership compounds, or exit infrastructure disappears, the window can narrow even if no acute crisis is visible.

This is the sense in which the paper uses hysteresis: the state reached after prolonged AI dependence may not be restored simply by removing the AI system. Skill, institutions, ownership, or social roles may have to be rebuilt.

Proposition 3 (Reversibility Principle) [OPEN]. The faster machine capability grows relative to human conversion, the more valuable it becomes to preserve reversible human and institutional alternatives before the recovery cost of lost routes rises.

9 AN URGENCY VECTOR INSTEAD OF A PANIC INDEX

Because Human Conversion is multidimensional, urgency should not be forced into one unvalidated scalar. Normalize machine growth and each human dimension within a declared study. Define a conversion-gap term [DEFINITION]

$$\Delta g_{j,t} = [\tilde{g}_{M,t} - \tilde{g}_{C,j,t}]_+. \quad (18)$$

Let $L_{j,t} \in [0, 1]$ be lock-in intensity, $S_{j,t}$ severity if the dimension deteriorates, and $\tau_{j,t}^{\text{rec}}$ expected recovery time. Then

$$U_{j,t} = \Delta g_{j,t} L_{j,t} S_{j,t} \tau_{j,t}^{\text{rec}}. \quad (19)$$

The Urgency Vector [DEFINITION] is

$$U_t = \langle U_{1,t}, \dots, U_{J,t} \rangle. \quad (20)$$

For precautionary reporting, a bottleneck summary may use

$$U_t^{\max} = \max_j U_{j,t}, \quad (21)$$

which prevents severe deterioration in one dimension from being averaged away by gains elsewhere. A weighted average is permissible only with explicit normative weights.

The logic is asymmetric. If acceleration is overestimated but safeguards preserve exit, skill, plural verification, and broad claims, social cost may be modest. If acceleration is underestimated and lock-in is real, late reconstruction may be much more expensive. This is an option-preservation argument under uncertainty, not a prediction of catastrophe [Arrow and Fisher, 1974, Fisher and Krutilla, 1974].

10 DISTRIBUTION: WHOSE POTENTIAL EXPANDS?

Average human conversion can conceal stratification. Highly skilled workers may use AI to widen alternatives and accelerate learning while novices use the same systems as substitutes for foundational reasoning. Recent experimental and conceptual work already points to heterogeneous learning effects and the difficulty of optimizing human learning relative to immediate accuracy [Bastani et al., 2025, Buçinca et al., 2024, Cash et al., 2026].

Therefore every population-level report should publish distributional readouts [DEFINITION]. For any dimension C_j :

$$C_{j,t}^{10}, \quad C_{j,t}^{50}, \quad C_{j,t}^{90} \quad (22)$$

and a simple spread

$$I_{j,t}^H = C_{j,t}^{90} - C_{j,t}^{10}. \quad (23)$$

A society in which elite Human Return and agency rise while the majority become dependent users is not “maximum human potential” merely because the mean rises.

Proposition 4 (Distributional conversion) [OPEN]. Broad human expansion requires that conversion gains reach a declared population distribution; aggregate productivity or mean capability gains are insufficient.

11 SIX MINIMUM SAFEGUARD INVARIANTS

The proposal does not prescribe one constitutional or economic system. It proposes six invariants that remain rational across a wide range of AI-speed forecasts because they preserve conversion and reversibility.

1. Human Return measurement. Systems that claim learning or human development should measure delayed unaided Concept reconstruction, Tool selection, Skill execution, and Alternative generation after decisive assistance is removed. Assisted performance alone is not evidence of durable human gain.

2. Independent resistance. AI candidate generation should not become the sole certifier of its own outputs. High-stakes inquiry requires target-domain evidence and partially independent critical routes.

3. Credible exit and interoperability. Provider count is not enough. Users and institutions should be able to switch, export, continue, or route around a failing or coercive gate at tolerable cost.

4. Convert automation dividends into durable claims and capacity. Transfers can preserve consumption today while machine-productive ownership continues to compound elsewhere. At least some productivity gains should create durable broad claims, time, learning capacity, or public/collective infrastructure rather than consumption alone.

5. Preserve social reproduction and critical unaided competence. Care, health, education, community, and selected non-AI critical skills should remain independently reproducible. Human productive necessity can fall while human social necessity remains.

6. Preserve contestability across economic, informational, and coercive power. Ownership, information mediation, and coercive authority should be measured separately. A society is most vulnerable when the same trajectory reduces exit across all three.

These are not bans on AI. Their purpose is to keep the conversion process corrigible.

12 A TIER-ONE EMPIRICAL PROGRAMME

The framework is designed to be attacked by rival models. No single study can test the whole architecture.

12.1 Longitudinal Human-AI learning experiments

Randomize assistance policies, not merely AI access. Compare direct-answer AI, friction-preserving tutoring, adversarial/counterargument AI, and no-AI controls. Measure assisted performance, delayed Human Return, transfer, error detection, metacognitive calibration, and independent route use. The core test is whether assistance optimized for current performance differs from assistance optimized for conversion.

12.2 Workplace conversion panels

Follow workers before, during, and after sustained AI adoption. Measure task productivity, wage/bargaining outcomes, skill retention after tool withdrawal, ability to detect tool failure, switching cost, and career mobility. A useful design compares occupations with similar AI productivity gains but different training and exit institutions.

12.3 Gate and exit audits

For compute, model access, cloud, payments, logistics, or information mediation, estimate counterfactual viability when a provider is removed. This distinguishes market share from functional gate power. The dependent variable is whether consequential activity can continue at declared cost, latency, quality, and legal constraints.

12.4 Cross-national conversion accounts

Extend national AI/productivity accounts with broad ownership of machine-capital income, labour centrality, cross-border AI rents, effective material claim, public compute/interoperability, social-reproduction capacity, and Human Return measures. This would test whether comparable AI adoption produces different human outcomes under different institutional conversion systems. The ILO-World Bank finding that developing economies may face “disruption without dividend” motivates this design [International Labour Organization and World Bank, 2026].

12.5 Reversibility estimation

Estimate recovery time and cost after dependence. Examples include retraining after skill decay, restoring offline/manual capacity, re-entering a market after a platform exit, rebuilding public expertise after outsourcing, or migrating standards after vendor lock-in. The Reversibility Principle survives only if these costs systematically rise with prolonged dependence.

13 RIVAL MODELS AND REMOVAL CONDITIONS

A theory proposal that cannot lose is only vocabulary. The following rival explanations must be compared explicitly:

- **Productivity-only model:** immediate output captures the relevant benefit.
- **Human-capital model:** conventional learning/skill measures explain all durable effects; Human Return adds nothing.
- **Capability/affordance model:** resources and formal capabilities fully explain option use; Live Possibility adds no information.
- **Competition model:** market concentration and switching cost are sufficient; conversion-gate and multidimensional power measures add no value.

- **Standard path-dependence model:** generic lock-in fully explains reversibility; human-specific conversion terms are redundant.

The Human Conversion architecture should be narrowed or removed under the following results:

1. delayed unaided performance is fully predicted by ordinary skill measures and never dissociates from assisted performance;
2. live-option measures add no predictive value beyond cost, preference, salience, and formal capability;
3. credible-exit and gate measures add no information beyond ordinary market concentration;
4. no dimension shows increasing recovery cost or path dependence under sustained AI reliance;
5. distributional conversion outcomes are fully explained by baseline income/education without AI-use interactions;
6. the vector architecture does not improve prediction, diagnosis, or intervention compared with simpler rivals.

14 WHAT THE PROPOSAL DOES NOT CLAIM

First, it does not claim that AI is intrinsically corrosive. The same technology can produce high or low conversion depending on design and institutions. Second, it does not claim that cognitive offloading is always harmful; strategic offloading can free resources for higher-level work. Third, it does not claim that employment must vanish. Labour centrality can persist through complementarity and new tasks. Fourth, it does not treat China as the cause of the global transition. China’s industrial-robot scale is a timing anchor for embodied automation, while U.S. and other firms remain central in frontier models, cloud, and compute. Fifth, it does not infer domination from concentration alone. Sixth, it does not forecast a date at which the Reversibility Window closes.

The proposal makes a narrower claim: where machine-side change is rapid and human/institutional restoration is potentially slow, waiting for definitive evidence of system-wide damage can itself destroy the options needed for low-cost correction.

15 DISCUSSION: FROM MAXIMUM ASSISTANCE TO MAXIMUM CONVERSION

The framework changes the design objective for human-AI collaboration. The usual objective is to maximize performance under assistance. The stronger objective is to maximize the quality of the human state that emerges after assistance. This implies a shift:

$$\begin{array}{l} \text{Maximum AI assistance} \longrightarrow \\ \text{Maximum durable human conversion.} \end{array} \quad (24)$$

In education, the goal is not to preserve every unaided task but to decide which foundational capacities must exist before higher-order offloading is safe. In work, the goal is not to block automation but to ensure that productivity gains produce

durable worker capability, mobility, or claims rather than only substitution. In science, AI should expand candidate space while evidence and independent criticism retain validation authority. In political economy, automation dividends should widen exit and agency rather than merely finance dependence.

The resulting division of labour is deliberately asymmetric. Machines can carry more search, recombination, translation, routine execution, and simulation. Humans need not compete at machine speed. Their strategic role is to retain problem ownership, normative judgment, responsibility, world-contact, corrigibility, and the capacity to refuse or redirect the system.

16 CONCLUSION

The defining competition of an AI-robotic century may not be human intelligence versus machine intelligence. It may be a race between two rates:

$$\begin{array}{l} \text{rate of machine capability growth} \\ \text{versus} \\ \text{rate of human conversion and institutional adaptation.} \end{array} \quad (25)$$

If machine capability rises while effective claim, exit, live possibility, corrigible agency, independent resistance, Human Return, and social reproduction also rise broadly, automation can release humans from unnecessary labour while expanding what they can understand, choose, and become. If machine capability rises faster than those conversion channels, the same productivity can produce a different trajectory: high assisted performance, high aggregate output, and growing human dependency.

The decisive policy window therefore lies before visible crisis. Skills can atrophy, ownership can compound, standards can harden, social institutions can disappear, and gate dependence can become normal. Under uncertainty, the rational response is not panic or prohibition. It is to preserve reversibility while measuring conversion.

The proposal can be compressed into one architecture:

$$\begin{array}{l} \text{Machine Capability} \rightarrow \text{Human Conversion Vector} \\ \rightarrow \text{Human Return/Agency/Exit} \\ \rightarrow \text{Reversibility or Lock-in} \\ \rightarrow \text{Next Institutional State.} \end{array} \quad (26)$$

The Human Conversion Imperative is therefore a design principle for an AI-rich world: do not ask only how much machines can do for humans now; measure how much capability, agency, ownership, and reversibility remain with humans afterward.

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